

Paper 1 - Set 05 - Solutions

Generated practice paper: answer key and worked solutions.

Answer key

Question	Answer	Topic
1	C	Differentiation
2	D	Quadratic functions
3	E	Ratio, proportion and percentages
4	F	Number, units and estimation
5	A	Graphs of functions
6	F	Straight lines
7	B	Polynomials, factor and remainder theorems
8	D	Probability
9	F	Simultaneous equations and inequalities
10	A	Polynomials, factor and remainder theorems
11	C	Triangles, radians and sectors
12	B	Quadratic functions
13	A	Graphs of functions
14	C	Integration
15	D	Functions and transformations
16	D	Pythagoras, mensuration and 3D
17	E	Trigonometric functions and equations
18	C	Straight lines
19	A	Exponentials and logarithms
20	A	Circles and circle theorems

Worked solutions

Question 1 DIF-10 • Differentiation • spec MM6.3

The curve $y = x^3 + px^2 + qx$ has stationary points at $x = 1$ and $x = 3$. Find q .

Answer: C

$$y' = 3x^2 + 2px + q = 3(x - 1)(x - 3) = 3x^2 - 12x + 9. \text{ So } q = 9 \text{ (and } p = -6\text{).}$$

Question 2 QUA-19 • Quadratic functions • spec MM1.3

The expression $x^2 + 6x + c$ can be written as $(x + p)^2 + q$. Given that $q = -c$, find the value of c .

Answer: D

$$(x + 3)^2 + (c - 9), \text{ so } q = c - 9 = -c, \text{ giving } 2c = 9 \text{ and } c = \frac{9}{2}.$$

Question 3 RAT-07 • Ratio, proportion and percentages • spec M3.5

A mixture contains sand and cement in the ratio 5 : 2. How much cement must be added to 35 kg of the mixture to make the ratio 5 : 3?

Answer: E

35 kg contains 25 kg sand and 10 kg cement. For 5 : 3 with 25 kg sand, cement must be 15 kg: add 5 kg.

Question 4 NUM-05 • Number, units and estimation • spec M2.8

Evaluate $\frac{(3 \times 10^4) \times (8 \times 10^{-2})}{6 \times 10^{-3}}$, giving your answer in standard form.

Answer: F

$$\frac{24 \times 10^2}{6 \times 10^{-3}} = 4 \times 10^5.$$

Question 5 GRA-09 • Graphs of functions • spec MM8.1

The graph of $y = \log_2 x$ is reflected in the line $y = x$. The resulting graph has equation

Answer: A

Reflection in $y = x$ gives the inverse function: $x = \log_2 y$, i.e. $y = 2^x$.

Question 6 LIN-02 • Straight lines • spec MM3.1

The points $A(1, 2)$, $B(5, 4)$ and $C(k, 8)$ are collinear. Find k .

Answer: F

$$\text{Gradient } AB = \frac{2}{4} = \frac{1}{2}. \text{ Gradient } AC = \frac{6}{k-1} = \frac{1}{2} \Rightarrow k - 1 = 12 \Rightarrow k = 13.$$

Question 7 POL-05 • Polynomials, factor and remainder theorems • spec MM1.6

The polynomial $p(x) = x^3 + ax^2 + bx + 4$ has $(x - 1)$ and $(x + 2)$ as factors. Find the third linear factor.

Answer: B

Write $p(x) = (x - 1)(x + 2)(x - r)$. The constant term is $(-1)(2)(-r) = 2r = 4$, so $r = 2$: the third factor is $(x - 2)$.

Question 8 PRO-01 · Probability · spec M7.7

A bag contains 4 red and 6 blue counters. Two counters are taken at random without replacement. Find the probability that they are different colours.

Answer: D

$$P(RB) + P(BR) = \frac{4}{10} \cdot \frac{6}{9} + \frac{6}{10} \cdot \frac{4}{9} = \frac{48}{90} = \frac{8}{15}.$$

Question 9 SIM-06 · Simultaneous equations and inequalities · spec MM1.4

Solve $xy = 6$ and $x - y = 1$. What is the sum of all possible values of $x + y$?

Answer: F

$x = y + 1$: $y^2 + y - 6 = 0 \Rightarrow (y + 3)(y - 2) = 0$. Solutions $(3, 2)$ with $x + y = 5$ and $(-2, -3)$ with $x + y = -5$. The sum is 0.

Question 10 POL-11 · Polynomials, factor and remainder theorems · spec MM1.6

Find the coefficient of x^2 in the expansion of $(x + 1)(x - 2)(2x + 3)$.

Answer: A

$(x + 1)(x - 2) = x^2 - x - 2$; multiply by $(2x + 3)$: the x^2 term is $3x^2 - 2x^2 = x^2$, coefficient 1.

Question 11 TRI-09 · Triangles, radians and sectors · spec MM4.1

Two sides of a triangle are 3 and 5 and the included angle is θ , where $\cos \theta = -\frac{1}{5}$. Find the length of the third side.

Answer: C

$$c^2 = 9 + 25 - 2 \cdot 3 \cdot 5 \cdot \left(-\frac{1}{5}\right) = 34 + 6 = 40.$$

Question 12 QUA-02 · Quadratic functions · spec MM1.3

By completing the square, find the minimum value of $2x^2 - 12x + 23$.

Answer: B

$$2x^2 - 12x + 23 = 2(x - 3)^2 - 18 + 23 = 2(x - 3)^2 + 5, \text{ so the minimum is 5 (at } x = 3\text{).}$$

Question 13 GRA-01 · Graphs of functions · spec MM8.6

How many times does the graph of $y = x^3 - x^2 - 6x$ cross or touch the x -axis?

Answer: A

$$x(x^2 - x - 6) = x(x - 3)(x + 2); \text{ three distinct real roots, so three crossings.}$$

Question 14 INT-12 · Integration · spec MM7.2

Find the value of $\int_1^8 x^{-\frac{2}{3}} dx$.

Answer: C

$$[3x^{1/3}]_1^8 = 3(2 - 1) = 3.$$

Question 15 FUN-09 · Functions and transformations · spec MM8.2

The graph of $y = \sin x$ is transformed to $y = 2\sin(3x) + 1$. What is the maximum value of the new function, and what is its period (in degrees)?

Answer: D

$\sin(3x)$ has period $360^\circ/3 = 120^\circ$. $2\sin(3x)$ ranges from -2 to 2 , so $2\sin(3x) + 1$ has maximum 3 .

Question 16 MEN-02 · Pythagoras, mensuration and 3D · spec M5.16

A cylinder has radius r and height h . Its volume is 54π and $h = 2r$. Find its total surface area.

Answer: D

$\pi r^2 \cdot 2r = 2\pi r^3 = 54\pi \Rightarrow r = 3, h = 6$. Surface area $= 2\pi r^2 + 2\pi rh = 18\pi + 36\pi = 54\pi$.

Question 17 TRG-02 · Trigonometric functions and equations · spec MM4.5

Solve $2\cos^2 x + 3\sin x - 3 = 0$ for $0^\circ \leq x \leq 360^\circ$. How many solutions are there?

Answer: E

$2(1 - \sin^2 x) + 3\sin x - 3 = 0 \Rightarrow 2\sin^2 x - 3\sin x + 1 = 0 \Rightarrow (2\sin x - 1)(\sin x - 1) = 0$. $\sin x = \frac{1}{2}$: $30^\circ, 150^\circ$; $\sin x = 1$: 90° . Three solutions.

Question 18 LIN-12 · Straight lines · spec MM3.1

The lines $y = mx + 4$ and $y = 3x - 2$ intersect at a point with positive x - and y -coordinates. Find the complete set of values of m .

Answer: C

$mx + 4 = 3x - 2 \Rightarrow x = \frac{6}{3-m}$, which is positive iff $m < 3$. Then $y = 3x - 2 > 0$ iff $x > \frac{2}{3}$, i.e. $\frac{6}{3-m} > \frac{2}{3}$; with $3 - m > 0$ this is $18 > 6 - 2m$, i.e. $m > -6$. So $-6 < m < 3$.

Question 19 LOG-06 · Exponentials and logarithms · spec MM5.1

The graph of $y = 2^x$ is translated 3 units in the positive x -direction. The resulting graph has equation

Answer: A

$y = 2^{x-3} = 2^{-3} \cdot 2^x = \frac{1}{8} \cdot 2^x$. (A translation of an exponential is also a stretch.)

Question 20 CIR-05 · Circles and circle theorems · spec MM3.2

The line $y = x + c$ is a tangent to the circle $(x - 1)^2 + (y - 2)^2 = 8$. Find the possible values of c .

Answer: A

The distance from $(1, 2)$ to $x - y + c = 0$ is $\frac{|1-2+c|}{\sqrt{2}} = \sqrt{8} = 2\sqrt{2}$, so $|c - 1| = 4$: $c = 5$ or $c = -3$.